

SOLVERS MATHEMATICS — CLASS 7

CHAPTER 8: WORKING WITH FRACTIONS

CHAPTER ASSESSMENT TEST PAPER

Time Allowed: 1 Hour | **Maximum Marks:** 40 Marks

SECTION A: OBJECTIVE TYPE QUESTIONS (10 Marks)

I. Multiple Choice Questions (1 Mark each)

1. A fraction whose numerator is greater than its denominator is called a/an:

[1 Mark]

- (a) Proper fraction
- (b) Improper fraction
- (c) Unit fraction
- (d) Equivalent fraction

2. The reciprocal of $2\frac{3}{5}$ is:

[1 Mark]

- (a) $\frac{5}{13}$
- (b) $\frac{13}{5}$
- (c) $2\frac{5}{3}$
- (d) $\frac{5}{6}$

3. The product of $\frac{4}{7} \times \frac{21}{16}$ in simplest form is:

[1 Mark]

- (a) $\frac{3}{4}$
- (b) $\frac{7}{4}$
- (c) $\frac{4}{3}$
- (d) $\frac{84}{112}$

4. Which of the following fractions is in simplest irreducible form?

[1 Mark]

- (a) $\frac{9}{21}$
- (b) $\frac{15}{25}$
- (c) $\frac{8}{15}$
- (d) $\frac{14}{49}$

5. What is the value of $\frac{3}{4} \div \frac{9}{8}$?

[1 Mark]

- (a) $\frac{2}{3}$
- (b) $\frac{3}{2}$
- (c) $\frac{27}{32}$
- (d) $\frac{1}{2}$

II. Fill in the Blanks (1 Mark each)

6. $\frac{3}{8} + \frac{5}{8} =$ _____.

[1 Mark]

7. To divide a fraction by another fraction, we multiply the first fraction by the _____ of the second fraction.

8. $\frac{2}{5}$ of 40 kg = _____ kg.

[1 Mark]

[1 Mark]

III. True or False (1 Mark each)

9. Multiplying a proper fraction by another proper fraction always yields a product smaller than each of the fractions.

10. The fraction $\frac{0}{5}$ has no reciprocal.

[1 Mark]

[1 Mark]

SECTION B: VERY SHORT ANSWER QUESTIONS (6 Marks)

11. Convert $\frac{43}{6}$ into a mixed fraction.

[2 Mark]

12. Find an equivalent fraction of $\frac{3}{7}$ having denominator 35.

[2 Mark]

13. Compare using $>$, $<$, or $=$:

[2 Mark]

$$\frac{5}{8} \quad \square \quad \frac{7}{12}$$

SECTION C: SHORT ANSWER QUESTIONS (12 Marks)

14. Evaluate step-by-step:

[3 Mark]

$$3\frac{1}{4} + 2\frac{2}{5} - 1\frac{1}{2}$$

15. A rectangular photo frame has length $12\frac{1}{2}$ cm and breadth $8\frac{2}{3}$ cm. Find:

(a) The perimeter of the frame.

(b) The area of the frame.

[3 Mark]

16. Evaluate:

[3 Mark]

$$\left(\frac{5}{14} \times \frac{21}{25} \right) \div \frac{9}{10}$$

17. A ribbon of length $15\frac{3}{4}$ m is cut into equal pieces, each of length $1\frac{3}{4}$ m. How many pieces are obtained?

[3 Mark]

SECTION D: LONG ANSWER QUESTIONS (8 Marks)

18. (a) In a library, $\frac{2}{5}$ of the books are in English, $\frac{1}{3}$ are in Hindi, and the remaining 480 books are in regional languages.

Find the total number of books in the library.

(b) How many books are in English, and how many are in Hindi?

[4 Mark]

19. (a) Illustrate the multiplication $\frac{2}{3} \times \frac{3}{4}$ using an area partition grid model and calculate the product in simplest form.

(b) Evaluate:

[4 Mark]

$$5\frac{1}{3} \div \left[2\frac{2}{3} + \left\{ 1\frac{1}{2} \times \left(\frac{4}{5} - \frac{2}{5} \right) \right\} \right]$$

SECTION E: CASE-STUDY / COMPETENCY-BASED QUESTION (4 Marks)

20. **Case Study: Community Kitchen Baking Recipe**

A community bakery prepares special artisan bread loaves. The standard recipe for one batch requires:

- $3\frac{1}{2}$ cups of whole wheat flour

- $\frac{3}{4}$ cup of warm milk

- $\frac{2}{3}$ cup of natural honey

On Saturday, the master baker decides to make $4\frac{1}{2}$ batches to supply a food festival.

(a) How much warm milk is required for all $4\frac{1}{2}$ batches?

[1 Mark]

(b) How much honey is required for all $4\frac{1}{2}$ batches?

[1 Mark]

(c) If the bakery has a 15-cup sack of wheat flour, is it sufficient to bake all $4\frac{1}{2}$ batches? Calculate the exact surplus or shortage.

[2 Mark]